

Lab 4 - Compiling and Building Software

Michelle Shaw, ACEnet CRC
5 November, 2010

Exercises

- compile some source using various compiler options:
 - default
 - turn off optimizations
 - macros: -fast
 - timing tests at each step
- build some packages using the 4-step build process on Linux:
 - fftw3
 - Open MPI (likely on your own time...)
- write makefiles for:
 - make-ex1
 - make-ex2
 - make-ex3

Compilers at ACEnet...

- if you are building on ACEnet, there are some things that are worth knowing:
 - default cc in configure scripts is usually cc (fortran: f77), which it expects to be GNU but is actually Sun studio 12 due to an ACEnet-wide setting in your path
 - there are three versions of GCC on our systems:
 - gcc -> gcc v4.1.2 (fortran: gfortran)
 - gcc34 -> gcc v3.4.6 (fortran: g77)
 - gcc43 -> gcc 4.3.2 (fortran: gfortran43)
 - why is it important? different versions of gcc can have slightly different behaviors
 - other compilers:
 - PGI: /opt/pgi/linux86-64 (default version is 8.0-6)
 - SS12: /opt/sun/sunstudio12

Compiling Some C and Fortran Codes

- let's get acquainted more with compilers and such before starting in on the build process...
- files in:
 - `/globalscratch/dshaw/CMSC6940/c_fort_src`
- build using:
 - default options with
 - `pgcc, cc, gcc`
 - `pgf77, f77, g77`
 - `pgf90, f90, gfortran`
 - same compiler set, try to set optimizations higher using the `-O` switch
 - compile format:
 - `pgcc xas_ave_20.c`
 - `pgcc xas_ave_20.c -o xas_ave_20.exe`

Compiling Some C and Fortran Codes

- any errors in the compile process?
- introduce errors into the code and see what compiler/linker errors you get
- NOTE: build_scwbox.f has an input file (build_scwbox.inp) which means you can run this code:
 - `./build_scwbox.exe < build_scwbox.inp > mystruct.txt`
- build and run it
- introduce an indexing error and see if you can get it to spit out runtime errors

Building using GNU build process...

- fftw:
 - copy from /globalscratch/dshaw to a directory of your choice
 - unpack the file:
 - `tar -zxvf fftw-3.2.2.tar.gz`
 - `cd fftw-3.2.2/`
 - check options in this configure script with:
 - `./configure --help`
 - make a record of which options you want and whether or not the defaults are suitable
 - NOTE: if you need specific libraries or compilers, you need to check to make sure they are available
 - make sure to change the install location with the `--prefix=...` option

Finding things on a Linux system...

- the build process may require special user-installed libraries, include files, etc. that may not be in standard locations
- may require you to reset the default compiler (usually GCC)
- standard library/include locations:
 - /usr/include
 - /usr/lib, /usr/lib64
 - /lib, /lib64
 - /usr/local/lib, /usr/local/lib64, /usr/local/include
- non-standard locations:
 - /opt (and subdirs)
 - /usr/local/<package>/lib, /usr/local/<package>/include
 - others...

Useful Linux Commands for Looking Around...

- some useful linux commands for looking around:
 - `ls -rlth /usr/lib64 | grep 'libc*'`
 - `ls -rlth /usr/bin/*gcc*`
 - `which cc`
 - `which pgcc`
 - `ls -rlth /usr/local/lib`
 - `ls -rlth /usr/local/include`
 - `echo $LD_LIBRARY_PATH`
 - `echo $PATH`

Building using the GNU build process...

- run the configure script with:
 - `./configure <your options>`
 - (completing this step may necessitate a coffee run... ;))
- watch for any errors or things you don't expect, i.e. wrong compiler choice, wrong flags, etc.
- does the configure complete? if not, what was the reported error?
- run make:
 - `make >& make.log &`
 - `tail -f make.log`
- run make check and make install:
 - `make check >& check.log & ; tail -f check.log`
 - `make install >& install.log & ; tail -f install.log`

Building using the GNU build process...

- ok, now try changing compiler flags and rebuilding
 - to do this you'll need to do the following:
 - make distclean
 - rm make.log check.log install.log
 - then you need to go through the configure process again to rebuild the makefiles
 - then the next 3 steps, i.e. make, make check, make install...
- change to a different compiler suite - any difference?

Intro to Make

- in /globalscratch/dshaw/CMSC6940/, there is a file called README_bldex
- follow instructions to unpack the examples
- inside each example directory, there is a README that should tell you what you need to do to build the source
- don't worry about running it, just get it to build